AMERICAN PSYCHOLOGICAL ASSOCIATION

Emotion

ISSN: 1528-3542

2026, Vol. 26, No. 3, 556–566
https://doi.org/10.1037/emo0001591

Developmental Changes in Youth Affect: A Within-Person Approach

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The transition from childhood to adolescence is a period of social–emotional reorganization involving changes in affect. Most research has examined developmental changes in between-person affect. Few studies have investigated developmental changes in associations between individual emotions and the structure of affective experience in youth across developmental age. This study used exploratory graph analysis to assess developmental changes in emotional complexity using the Positive and Negative Affect Schedule administered at three time points from 2007 to 2013 in a three-cohort, accelerated longitudinal design spanning Grades 3 through 12 ($N = 682$): late childhood, $M_{age} = 9.39$, $SD = 0.53$; early adolescence, $M_{age} = 11.80$, $SD = 0.67$; and middle adolescence, $M_{age} = 14.60$, $SD = 0.60$. Decreases in edge density and entropy and increases in R^2 were identified across development. In contrast, nonlinear shifts were found for the number of negative edges between affective dimensions and mean absolute error and possible shifts in dimensionality. Results suggest that global network metrics support decreases in emotional complexity from childhood through adolescence, though other indices suggest distinct patterns of change. Implications for research and study limitations are discussed.

Keywords: positive affect, negative affect, youth development, exploratory graph analysis

Supplemental materials: https://doi.org/10.1037/emo0001591.supp

Affective health is a central facet of well-being in youth, as trait-like positive and negative affect have been associated with a host of outcomes in adolescent development including relationship quality (e.g., Griffith, Young, & Hankin, 2021), physical health (e.g., Pressman et al., 2019), occupational performance and success (e.g., Kaplan et al., 2009; Lyubomirsky et al., 2005), and psychopathology (e.g., Clark & Watson, 1991; Joiner et al., 1996). Researchers have shown that youth experience normative decreases in trait positive affect (PA) and increases in negative affect (NA) during the transition from childhood to adolescence, a period of significant social–emotional reorganization (e.g., Coe-Odess et al., 2019; Griffith, Clark, et al., 2021; Guyer et al., 2016; Larson et al., 1980; Rapee et al., 2019). However, additional research is needed to understand developmental changes in the *structure* of affect across this developmental period. That is, much is left to be learned regarding how interconnections between NA and PA evolve from childhood through adolescence, including how emotional complexity changes across development.

For decades, factor analytic researchers have endeavored to clarify the structure of affect, with results generally yielding support for a two-factor model, featuring PA and NA (Crawford & Henry, 2004; Rush & Hofer, 2014; Tellegen et al., 1999; Watson et al., 1988). Affective scientists have long recognized that affect is structured hierarchically, with discrete emotions contributing unique and meaningful variance to outcomes of interest (Izard, 1993; Lench et al., 2011; Tellegen et al., 1999). Developmental researchers have predominantly focused on

This article was published Online First October 6, 2025.
Kalina Michalska served as action editor.
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The study data are available on the Open Science Framework at https://osf.io/gtq94/. The research reported in this article was supported by grants from the National Institute of Mental Health, National Institutes of Health, to Jami F. Young (Grant R01MH077178) and to Benjamin L. Hankin (Grant R01MH077195). George Abitante was supported by the National Institute of Mental Health, National Institutes of Health (Grant NIH T32 MH018921), during the completion of this work.

George Abitante played a lead role in writing–original draft and writing–review and editing and an equal role in conceptualization, formal analysis, and visualization. Julianne M. Griffith played a supporting role in writing–original draft and writing–review and editing and an equal role in conceptualization. Alexander P. Christensen played a supporting role in writing–review and editing and an equal role in formal analysis and visualization. David Cole played a supporting role in writing–original draft and writing–review and editing and an equal role in conceptualization. Jami F. Young played a supporting role in writing–review and editing and an equal role in data curation and funding acquisition. Benjamin L. Hankin played a supporting role in writing–original draft and writing–review and editing and an equal role in conceptualization, data curation, and funding acquisition.

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overarching PA and NA dimensions, with relatively less research on the complex interrelations among the individual emotions that underlie each of these constructs. This body of work has generated rich insights into the affective ecology of adolescent development. However, emerging evidence suggests that distinct negative and positive emotions may demonstrate differing patterns of development and associations during adolescence (Bailen et al., 2019), highlighting the need for more research examining structural characteristics of youth emotional experience.

Research on youth emotional development often has focused more on between-person than within-person relations among positive and negative emotions. This is notable, as factor analysis indicates that structural qualities of affect (e.g., PA and NA correlations) may differ across within- and between-person levels (Rush & Hofer, 2014). Interest in understanding within- and between-person relations has led to the application of increasingly advanced statistical approaches to affect (Lange & Zickfeld, 2021; Molenaar, 2004). One such approach, network analysis, has been fruitfully applied to the structure of affect, including the cross-sectional, between-person interrelations of discrete negative and positive emotions among adults (Flores-Kanter et al., 2021; Lange & Zickfeld, 2021). However, research is needed to advance knowledge of *change* in distinct positive and negative emotions from childhood through adolescence (Flores-Kanter et al., 2021; Funkhouser et al., 2021).

Emotion Complexity in Adolescence

Numerous studies have characterized developmental changes in affective experiences in childhood and adolescence (e.g., Griffith, Clark, et al., 2021; Larson et al., 2002; Moneta et al., 2001; Olino et al., 2011). Less is known, however, about developmental changes in emotional complexity, a term that encompasses numerous definitions and operationalizations and refers to the simultaneity and granularity of emotional experiences (O'Toole et al., 2020). Emotion complexity has been operationalized in myriad ways, including the intraclass correlation, associations between PA and NA, and the number of components identified using principal component analysis (e.g., Brose, de Roover, et al., 2015; Dejonckheere et al., 2018; O'Toole et al., 2020). The intraclass correlation is thought to represent emotion differentiation, whereas associations between PA and NA are thought to reflect emotion covariation. The number of components identified may best reflect emotion differentiation, as suggested by Brose, de Roover, et al. (2015). Prior work suggests that heightened emotional complexity relates to improved psychological well-being, including reductions in binge eating and substance use and better prognosis in adolescence (Schreuder et al., 2022; Seah & Coifman, 2022).

There is some evidence of developmental changes in emotional complexity during adolescence from studies using the intraclass correlation. Nook et al. (2018) studied a cross-sectional sample of youth aged 5–25 and found emotional complexity was higher in childhood and young adulthood compared to adolescence. An ecological momentary assessment study of youth aged 14–17 found linear decreases in negative emotional complexity with age (Starr et al., 2020). These studies provide important foundational evidence for differences in emotional complexity by age, but additional studies are needed to clarify developmental trends in additional facets of emotional complexity. Recent innovations in statistical approaches suggest novel avenues to investigate such developmental changes in emotional complexity.

Exploratory Graph Analysis

Network analysis is a statistical approach that enables analysis of associations between distinct emotions and therefore yields insight into youth emotional complexity. Network analyses represent individual emotions as "nodes" that comprise an interconnected network of associations. Associations between nodes are referred to as "edges," which can vary in strength and direction and represent partial correlations between emotions while accounting for all other nodes in the network. To date, network analysis has been used to study interconnections between diverse aspects of youth functioning and symptoms of psychopathology (e.g., Bringmann et al., 2016; Campbell & Osborn, 2021; Funkhouser et al., 2021; Kuranova et al., 2021). The promise of this approach in illuminating novel aspects of youth's complex, multidimensional emotional development has not been realized, however.

Exploratory graph analysis (EGA; Golino & Epskamp, 2017; Golino et al., 2020) is a recent development in network analysis that can extend our understanding of youth emotional development. EGA allows researchers to assess the dimensionality of temporal, multivariate data. Dimensions in a network are represented by communities or sets of interconnected nodes that emerge from the relations between the nodes (Golino & Epskamp, 2017). Networks requiring a larger number of dimensions reflect greater nuance among the components of that network.

Dynamic EGA was developed to assess the dimensionality of within-person psychological phenomena using first-order derivatives (Golino et al., 2022). These derivatives represent change in variables over time (e.g., increasing happiness). Positive edges between nodes indicate that across people, there are mutual increases or decreases in within-person changes, whereas negative edges indicate opposing fluctuations. Several aspects of EGA networks can be used to assess developmental changes in affective complexity that align with prior work. First, increases in the number of dimensions in a network suggest increasing complexity. Such changes align with prior work examining the number of components representing affect and thus emotion differentiation. Second, stronger associations between PA and NA dimensions indicate greater emotion covariation and lower complexity. Third, increases in the density of edges in the network may reflect increasing complexity because the network has greater influence over individual nodes when more edges exist.

In addition to these metrics, three other indices of complexity are available using EGA. Two indices involve node predictability, or the extent to which the variability of a node can be explained by other nodes (Haslbeck & Waldorp, 2018). Node predictability can be assessed using R^2, which assesses the proportion of variance in a node that is explained by other nodes, and the mean absolute error (MAE), which represents the average magnitude of errors predicting an observation's value for a node using the other nodes in the network. Increasing R^2 values reflect decreasing complexity, since more of a node's variance can be explained by the network, whereas increasing MAE suggests greater complexity because it is harder to predict a node's value using the rest of the network. Additionally, an information-theory metric, von Neumann entropy, assesses how (dis)organized a network is, with higher values reflecting greater disorganization or network complexity (De Domenico et al., 2015; Golino et al., 2025). EGA therefore provides us with several novel indices of emotional complexity that allow us to assess developmental changes in the complexity of affect from childhood through adolescence.

Limitations of Previous Research

Prior studies have generated invaluable insights into youth affective experience and development. Nevertheless, key gaps remain. First, the predominance of research on child and adolescent emotional development has relied on summary score approaches that bin discrete emotions into broad "positive" and "negative" affective categories. This conceptual approach is strongly supported by decades of factor analytic work (e.g., Tellegen et al., 1999; Watson et al., 1988), but knowledge generated using this analytic strategy is incomplete (e.g., Tellegen et al., 1999). For example, variation in the valence (positive/negative) *and* arousal (high/low) of different emotions produces distinct patterns of association with salient outcomes of interest (Domachowska et al., 2016; Gable & Harmon-Jones, 2010; Gilbert, 2012; McManus et al., 2019; Posner et al., 2005). Moreover, the density of emotional networks (i.e., degree of emotion differentiation) has been found to predict unique variance in key developmental outcomes (e.g., Seah & Coifman, 2022). Thus, research using more granular approaches to describe the pattern and strength of interconnections between distinct positive and negative emotions from childhood through adolescence is needed to enrich our understanding of youth affective experience.

Second, most studies to date of youth emotional experience, and meta-emotional processes like emotional complexity, have focused on between-person individual differences. Theories of adjustment and adaptation across the adolescent transition, however, rely on *within-person* patterns of association, which may not align with between-person findings (Brose, Voelkle, et al., 2015; Curran & Bauer, 2011; Hamaker et al., 2007). For instance, prior between-person studies of PA and NA often identify relatively orthogonal trait PA and NA factors at the between-subject level, whereas studies examining within-person correlations find small to moderate negative associations (e.g., Bleidorn & Peters, 2011; Brose, Voelkle, et al., 2015; Laurent et al., 1999; Merz & Roesch, 2011; Watson et al., 1988; Yamasaki et al., 2007).

Third, the ways in which development and change in one emotion (e.g., fear) are related to change in another emotion (e.g., sadness) are unclear. This is consequential, as identifying such nuanced associations informs the design of optimally targeted preventive interventions that promote youth affective well-being. Studies utilizing methods that assess and model changes in affect over time are essential to obtain a robust understanding of developmental changes in affect.

The Present Study

The transition from childhood to adolescence is a period of normative change across domains and units of analysis, including changes in mean-level affective experience. Findings from cross-sectional work indicate that other salient aspects of emotional experience, including patterns of interconnections between individual emotions, also likely undergo normative change during this time. However, prospective, repeated-measures research using advanced analytic designs is needed to clarify patterns of change in these granular aspects of youth affective experience. The present study uses EGA to address these gaps in the existing literature and answer two central questions. First, to what extent do changes in individual emotions predict each other across adolescence? Second, how do these individual emotions change together across time among different age groups in childhood and adolescence? Given the scarcity of previous research in this area, we did not make a priori hypotheses; this study was conceptualized as exploratory and descriptive rather than confirmatory in nature.

Method

Participants

Participants were 682 youth recruited in third ($N = 209$), sixth ($N = 249$), and ninth ($N = 224$) grade cohorts (age 8–16 at baseline, $M_{age} = 11.84$, $SD_{age} = 2.41$, 56% female; see Tables 1 and 2). Data collection occurred from 2007 until 2013. Inclusion criteria were English language fluency and absence of autism or psychotic disorder diagnosis. Additionally, to ensure participants could complete the battery of questionnaires, parents were asked whether their child had an intellectual disability or diagnosis of mental retardation. Children whose parents responded affirmatively were excluded from the study. Sample demographics were approximately representative of the ethnic and racial characteristics of the U.S. population (67.9% Caucasian, 11.3% African American, 9.1% Asian/Pacific Islander, 6.0% Multiracial, 4.8% identifying as Other, 0.9% American Indian/Alaskan Native, with 12.8% identifying as Hispanic/Latinx). All procedures were approved by the Institutional Review Board. Informed consent was obtained from all participating caregivers, and assent was obtained from all participating youth. Youth were invited to the laboratory to complete a battery of measures every 18 months for 3 years. Participants completed the Positive and Negative Affect Scale for Children (PANAS-C) at each time point, yielding three assessment points per participant or seven total assessment points spanning third to 12th grade using an accelerated longitudinal cohort design. We refer to cohorts as follows: late childhood (third grade at baseline), early adolescence (sixth grade at baseline), and middle adolescence (ninth grade at baseline). Additional details regarding sampling procedures and participant characteristics are described in Hankin et al. (2015).

Measures

Positive and Negative Affect

Trait affect was assessed every 18 months using youth self-report on the PANAS-C (Laurent et al., 1999). The PANAS-C is a reliable

Table 1
Summary Statistics and Correlation Matrices for Each Cohort

Cohort variable	*M* (*SD*)	1	2	3	4
Late childhood (Grade 3)					
1. Age	9.39 (0.53)	—			
2. Gender (female)	47.22%	.03	—		
3. Mean PA	3.74 (1.31)	.03	.14	—	
4. Mean NA	1.85 (1.14)	−.22	.25*	.32*	—
Early adolescence (Grade 6)					
1. Age	11.80 (0.67)	—			
2. Gender (female)	58.88%	.02	—		
3. Mean PA	3.74 (1.05)	−.03	.13	—	
4. Mean NA	1.73 (0.93)	.04	.16*	−.20*	—
Middle adolescence (Grade 9)					
1. Age	14.60 (0.60)	—			
2. Gender (female)	58.55%	−.14	—		
3. Mean PA	3.65 (1.04)	.01	.01	—	
4. Mean NA	1.99 (1.10)	−.01	.22*	−.11	—

Note. PA = positive affect; NA = negative affect.
* $p < .05$.

Table 2
Demographic Information

Demographic characteristic	Total sample (*N* = 682) *M* (*SD*)	Analytic sample (*n* = 394) *M* (*SD*)
Age	11.84 (2.41)	12.44 (2.04)
Household income	$99,095 ($80,070)	$103,466 ($76,806)
Demographic characteristic	***N* (%)**	***N* (%)**
Gender		
Male	301 (44.1)	171 (43.4)
Female	381 (55.9)	223 (56.6)
Grade		
3rd	209 (30.6)	72 (18.3)
6th	249 (36.5)	170 (43.1)
9th	224 (32.8)	152 (38.6)
Race		
American Indian/Alaskan Native	6 (0.9)	2 (0.5)
Asian/Pacific Islander	62 (9.1)	36 (9.1)
African American/Black	77 (11.3)	35 (8.9)
Caucasian	463 (67.9)	286 (72.6)
Middle Eastern	0 (0)	0 (0)
Other	33 (4.8)	13 (3.3)
Multiracial	41 (6.0)	22 (5.6)
Ethnicity		
Hispanic/Latinx	87 (12.8)	39 (9.9)
Not Hispanic/Latinx	595 (87.2)	355 (90.1)
Parent education level		
8th grade or less	3 (0.4)	0 (0)
Some high school	6 (0.9)	5 (1.3)
Finished high school	36 (5.3)	18 (4.6)
Completed GED	11 (1.6)	2 (0.5)
Vocational/trade/business school	37 (5.4)	17 (4.3)
Some college/2-year degree	196 (28.7)	100 (25.4)
Finished 4-year degree	194 (28.4)	123 (31.2)
Master's degree or equivalent	134 (19.6)	93 (23.6)
Other advanced degree	62 (9.1)	35 (8.9)
Food stamp receipt		
Yes	46 (6.7)	18 (4.6)
No	634 (93.0)	375 (95.4)

Note. Not all participants responded to each question, resulting in some counts that do not sum to the total/analytic sample size. GED = General Educational Development.

and commonly used questionnaire measure assessing youths' experience of 27 discrete emotion states (e.g., "interested," "sad," "excited") on a 5-point Likert scale from 1 (*very slightly or not at all*) to 5 (*extremely*). The PA subscale consists of 12 items assessing youths' experience of such positive emotion states as "cheerful," "delighted," and "calm." The NA subscale comprises an analogous 15 items assessing youths' feelings of such emotion states as "frightened," "ashamed," and "upset." Participants rated their affect during the past week. The PA and NA subscales of the PANAS-C demonstrate strong psychometric properties among adolescent samples and evidence good convergent and discriminant validity in both clinical (Hughes & Kendall, 2009) and community samples (Laurent et al., 1999). Overall, PA and NA subscales demonstrated good reliability at all assessment points in the present sample (α = .86–.90 and .86–.91 for PA and NA, respectively). Scale authors suggest that affect adjectives may not be well understood by children prior to achieving a Grade 4 reading level, so PANAS-C data were not included in analyses for participants prior to age 9 (Laurent et al., 1999). This resulted in the removal of 57 participants younger than age 9 at baseline.

Transparency and Openness

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study. Sample size for the initial study was informed by an a priori power analysis. The main purpose was to have sufficient power to detect small effects (effect size = .10) regarding prospective relations between depression and anxiety symptoms and various cognitive, interpersonal, and biogenetic predictors. Children were in third, sixth, or ninth grade at the time of study recruitment. As this study was observational, no manipulations were used. All study measures are included in the attachment. For secondary analysis, power analytic methods have not yet been developed for exploratory network analysis (but see Constantin et al., 2023). However, we note that prior studies using network approaches have used sample sizes comparable to ours (e.g., Park & Kim, 2020; Ruan et al., 2022).

Raw data cannot be shared because study participants were not offered open data sharing as an option during the consenting process, as the study began recruitment and data collection in 2007 before present transparency and openness practices. Additionally, university lawyers have advised that raw data cannot be openly shared due to the consenting process used when the study was active. Deidentified data may be available upon reasonable request to the lead author or Drs. Hankin or Young. Likewise, study documents and protocols are available upon request.

This study's design and hypotheses were preregistered after data had been collected but before analyses were undertaken, emphasizing assessing change within cohorts over time and comparisons of networks across cohorts (https://osf.io/93qet). After further consideration, and after the development of new analytic approaches that enabled the investigation of novel, exciting variations on our initial research question, our aims evolved to center upon the previously described exploratory analysis of changes in affective network structure across developmental cohorts. Although the details of the analytic approach have changed, we conceptualize the work reported herein as consistent with the initial aims described in the preregistration. Moreover, we continue to conceptualize the present work as primarily descriptive and exploratory. Analysis code is available at https://osf.io/gtq94/.

Analytic Plan

Estimating Change

To estimate within-person change over time, we computed derivatives of each variable across time points for every participant. Derivatives can capture different dynamics, including nonlinear relationships, across time. Derivatives were estimated using generalized local linear approximation (Deboeck et al., 2009), which uses a time-delay embedding matrix to assess change in each variable across time. A separate weight matrix was estimated using the time between successive observations in each time series, the number of embedded dimensions, and the order of the derivative of interest. For this study, the first-order derivatives were used to capture the rate of change of each variable across time. When these derivatives are correlated, the dynamics of how variables change together across time are captured (Golino et al., 2022). We used an embedding of three (number of waves), resulting in a single first-order derivative value for each variable for each participant. The final data matrix was a participant-by-variable matrix, identical to a cross-sectional matrix, which captured the rate of change for each variable rather than a single measurement point. These rate-of-change matrices were computed using only participants with complete data for each cohort. Network loadings were computed following up-to-date procedures (Christensen et al., 2024).

Reducing Redundancy

Before proceeding with the network analysis, we first submitted these rate-of-change matrices to unique variable analysis (Christensen et al., 2023) to identify local dependence between items, removing items exhibiting large violations. Redundancy (local dependence) can strongly bias the dimensionality and parameter estimates (e.g., edges, centrality). Flores-Kanter et al. (2021), for example, examined the cross-sectional dimensionality of the 20-item PANAS and found that redundancies led to a minor factor, resulting in three dimensions rather than the theoretical two dimensions. Further, network theory presupposes that components of a network are *unique* (i.e., causally autonomous), so redundancies in networks should be handled (Christensen et al., 2020; Cramer et al., 2012).

Dimensionality

After removing redundant items, we implemented EGA, a network analytic approach to identify dimensions in multivariate data. This approach estimates a Gaussian graphical model (Lauritzen, 1996) in which nodes (circles) represent variables and edges (lines) represent partial correlations. To estimate the Gaussian graphical model, the graphical least absolute shrinkage and selection operator (Friedman et al., 2008) was used. The graphical least absolute shrinkage and selection operator procedure was used to determine how many edges to include in the network based on two hyperparameters: lambda and gamma. For all networks, we used the default argument of lambda.min.ratio = 0.1 in EGAnet. To determine how complex the model should be, the extended Bayesian information criterion (Chen & Chen, 2008) was implemented with a hyperparameter gamma, which determines whether complex models are preferred. Last, to estimate whether nodes belong to distinct communities, the Walktrap community detection algorithm was used (Pons & Latapy, 2006). The Walktrap algorithm uses "random walks" to identify the content and number of dimensions in the network. Using an analytic solution, the network was converted into a probability transition matrix that represents the long-run expected average of the random walks.

Dimensionality Robustness

We performed bootstrap EGA using the parametric approach, which generates data from a multivariate normal distribution, with 1,000 iterations to assess the stability of dimensions identified in our networks (Christensen & Golino, 2021). Item stability, or the proportion of bootstrap samples that a node is replicated in the initial EGA communities, is considered robust when proportions are around or greater than .75 (Christensen & Golino, 2021). Similarly, dimension stability, or the proportion of bootstrap samples that each initial EGA community is replicated exactly, is considered robust when proportions are around or greater than .75.

Node Predictability

To determine each node's contribution to the network across time, we computed a variant of node predictability (Haslbeck & Waldorp, 2018) to determine how well each node could be predicted by its connections to the rest of the network. Node predictability leverages the equivalence between partial correlations and beta coefficients of linear regression to produce estimated values for each variable (Guttman, 1953). Variable-wise predictions can be evaluated for their "predictability," offering a straightforward metric to infer a node's connectivity in the network. Putting this information into context, predictability is the extent that a node can be predicted based on its network connectivity alone. Remaining predictability (i.e., if the accuracy is not 1) means that there is residual variance that could be explained by variables outside of the network (whether in the environment, other emotions, or internal conditions of the system). Using this approach, we obtained R^2 and MAE as indices of complexity. Interpretation of R^2 and MAE is identical in network models as in other analyses.

Von Neumann Entropy

We also computed von Neumann entropy for each cohort's network. The von Neumann entropy captures the disorder or uncertainty present in a network using its spectral properties (i.e., eigenvalues; De Domenico et al., 2015). Larger values indicate greater disorder and thus greater emotional complexity, whereas smaller values indicate less disorder or lower complexity. Since von Neumann entropy is a relative measure, there is no statistical test for comparing these values, so we used permutation tests to compare the cohorts.

All analyses were completed using the EGAnet package (Version 2.0.6; Golino & Christensen, 2024) in R Version 4.4.1 (R Core Team, 2024). Visualizations were created using ggplot2 (Version 3.51, Wickham, 2016), GGally (Version 2.2.1, Schloerke et al., 2024), and ggpubr (Version 0.6.0, Kassambara, 2023).

Results

Unique Variable Analysis

There were multiple pairs of items exhibiting large to very large redundancy (values > .30). Across cohorts, the items identified as redundant varied, so we only removed items identified as redundant for all cohorts. We identified four pairs of items exhibiting large to very large redundancy: Calm-Jittery, Happy-Nervous, Guilty-Scared, and Blue-Lively. Because many of these items represented distinct affective dimensions, we did not merge them, as is often done in redundancy analyses. Instead, using bootEGA, we assessed each item's stability and removed items with the lowest average stability across cohorts. In this way, we removed Calm, Nervous, Guilty, and Lively. All following analyses examined the PANAS-C excluding these items.

EGA

We then used the remaining variables to estimate networks for each cohort across panel waves using first-order derivatives. Figure 1 displays a comparison of longitudinal networks for each cohort (using the Grade 3 network as a template), in which green edges represent positive associations and red edges represent negative associations. These edges represent partial correlations between nodes, controlling for associations between all other nodes. Consequently, positive edges between PA nodes mean they are changing together, and negative edges between PA and NA nodes indicate that one increases while the other decreases.

Bootstrap Results

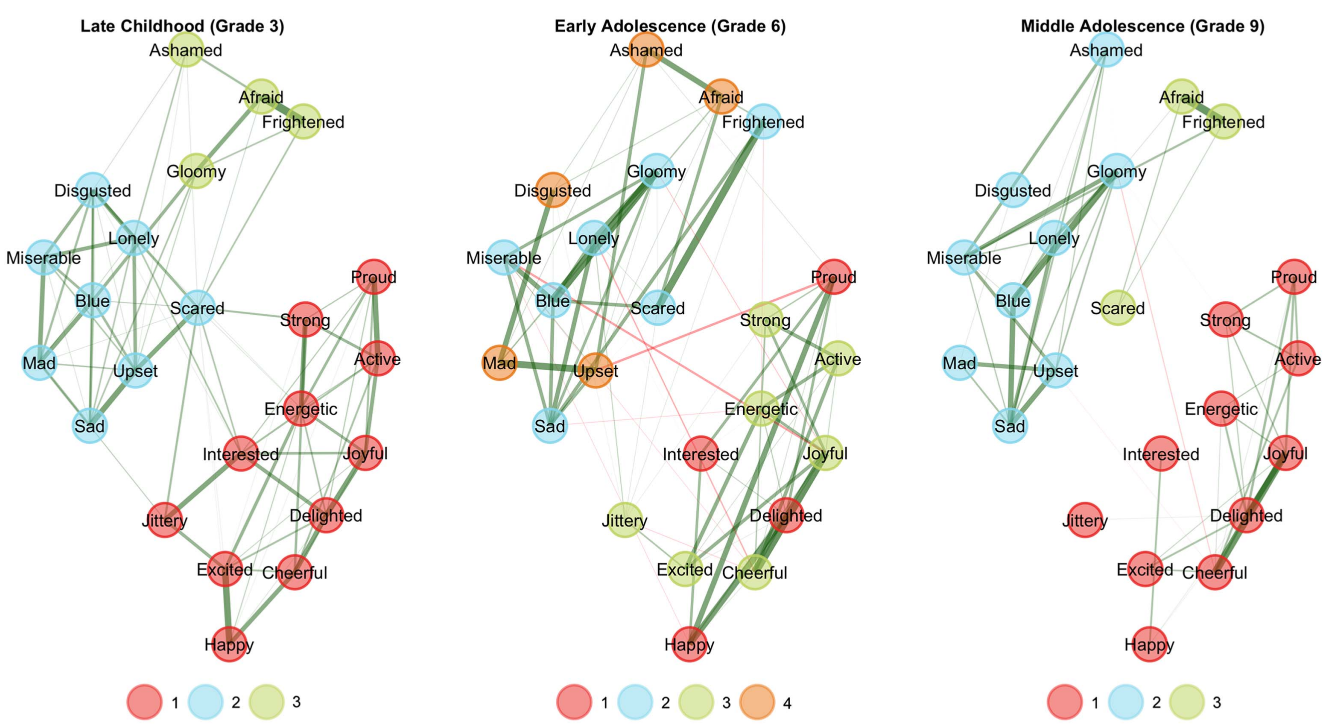
We used bootEGA to assess the stability of the dimensions identified in our empirical samples (see Table 3). We identified different numbers of dimensions for each cohort, with three dimensions for late childhood, four for early adolescence, and three for middle adolescence. Although the networks for late childhood and middle adolescence exhibited more replications for four dimensions, these were due to a single dimension identified in late childhood (Jittery) and a minor factor in middle adolescence (Happy, Interested). Consequently, three dimensions were estimated for these cohorts. Stability of 75% and higher is considered robust (Christensen & Golino, 2021). For late childhood, we found that 34% of iterations identified a three-dimension solution and 18 items exhibited good stability, except for Gloomy (68%), Scared (59%), Interested (56%), Ashamed (42%), and Jittery (35%). For early adolescence, we identified four dimensions most often (47%). Twenty items exhibited good stability, except for Scared and Frightened (52%) and Jittery (45%). For middle adolescence, we identified three dimensions 30% of the time and 20 items exhibited stability above 75%, with Interested and Disgusted (66%) and Happy (65%) exhibiting lower stability.

We identified important consistencies and differences across these networks. First, nodes were placed in dimensions consistent with PA and NA. The additional dimensions we identified represent facets, or subdimensions, of broad PA and NA dimensions. For instance, in each cohort, NA was represented by two dimensions, whereas PA was represented by one dimension in late childhood and middle adolescence, but two dimensions in early adolescence. Thus, the greater number of dimensions we identified appear to bin along theoretical PA and NA dimensions. Next, we observed decreases in edge density from late childhood through middle adolescence, suggesting decreasing emotional complexity (see Table 4). Last, we evaluated the interdependence of PA and NA dimensions by examining the number of negative edges between dimensions representing facets of PA and NA dimensions. We found fewer negative edges between PA and NA dimensions in late childhood (0) and middle adolescence (3) compared to early adolescence (9). These results suggest potential nonlinear shifts, in which PA and NA dimensions are relatively orthogonal in late childhood and middle adolescence and more inversely related in early adolescence.

Predictability Results

We then assessed the predictability of items (see Table 4; item-level results presented in Supplemental Table S1). For late childhood,

Figure 1
Networks With Identified Dimensions for Each Cohort



Note. See the online article for the color version of this figure.

Table 3
Estimated Number of Dimensions for Each Cohort

Cohort	Number of dimensions					
	One	Two	Three	Four	Five	Six
Late childhood (Grade 3)	0	<0.01	**0.34**	0.57	0.08	<0.01
Early adolescence (Grade 6)	0	0	0.03	**0.47**	0.46	<0.01
Middle adolescence (Grade 9)	0	0	**0.30**	0.62	0.09	<0.01

Note. Values in bold indicate the selected number of estimated dimensions. For late childhood, although four dimensions were identified often, this was due to a single-item dimension (Jittery), which exploratory graph analysis automatically removed. Similarly, for middle adolescence, the fourth dimension was a minor factor comprised of two items (Happy, Interested), so three dimensions better represented this cohort as well.

R^2 ranged from .187 (Jittery) to .569 (Delighted), with an average of .451, whereas MAE ranged from .338 (Miserable) to .903 (Jittery), with an average of .544. For early adolescence, R^2 ranged from .125 (Jittery) to .738 (Blue), with an average of .502, whereas MAE ranged from .273 (Miserable) to .759 (Jittery), with an average of .487. For middle adolescence, R^2 ranged from .201 (Interested) to .688 (Gloomy), with an average of .527, whereas MAE ranged from .371 (Miserable) to .877 (Jittery), with an average of .574. Additionally, we observed three NA values for R^2 in middle adolescence, reflecting insufficient variability in the predictors due to sparse connections (i.e., β coefficients of 0) with the target node. These results suggest that across cohorts, items varied in their predictability, suggesting some items (e.g., Jittery) had relatively little explained variance, whereas others (e.g., Blue, Gloomy) had considerable variance accounted for by other variables in the network.

Von Neumann Entropy

Last, we examined changes in von Neumann entropy between cohorts. Entropy shows how much uncertainty there is in the data, and higher values are thought to reflect greater complexity. Entropy declined from late childhood (4.28) to early adolescence (4.16) and middle adolescence (3.89). Permutation tests suggested significant decreases from late childhood to middle adolescence ($p = .012$) and from early to middle adolescence ($p = .040$) and a nonsignificant difference between late childhood and early adolescence ($p = .416$).

Discussion

In summary, we used EGA to identify possible developmental changes in affect across adolescence. We obtained results suggesting disparate patterns of findings. First, we identified linear decreases in edge density and entropy and linear increases in R^2, suggesting decreases in emotional complexity from childhood through adolescence. Second, we found nonlinear changes in the orthogonality of PA and NA and MAE, suggesting shifts in complexity depending on developmental period. Third, we observed shifts in network dimensionality from childhood through adolescence; however, several factors suggested caution is warranted in interpreting these changes. We discuss implications of our findings for understanding developmental changes in the structure of emotional change from childhood through adolescence.

First, we observed decreases in edge density and von Neumann entropy and increases in R^2, suggesting declines in emotional complexity from childhood through adolescence. Qualitatively, these shifts indicate reductions in the connections between emotions, suggesting shifts from more interconnected to independent change in emotions. Edge density and entropy represent the strongest global indices of complexity that we examined and provide complementary findings. When nodes exhibit fewer edges, it may be easier to explain variance in a node because it is connected to fewer, more strongly associated nodes. Additionally, as the number of edges decreases, we expect complexity to decrease because there are fewer ways for nodes to be influenced by other nodes in the network. These results support findings from prior studies indicating decreases in emotion differentiation in adolescence (Nook et al., 2018; Starr et al., 2020).

Second, we identified nonlinear changes in the number of negative edges between PA and NA dimensions and in MAE, suggesting distinct shifts from childhood through adolescence. These results may reflect developmental changes in the interdependence of PA and NA dimensions, and emotions more generally, from childhood through adolescence. Much like findings from McElroy et al. (2018) suggesting increasing connectivity among depressive symptoms from childhood to early adolescence, our results suggest increasing negative connections between PA and NA from late childhood into early adolescence. These negative edges may reflect greater reciprocal effects of affect dimensions in adolescence, such that depressed mood leads to anhedonia, or vice versa. Our findings also align with work suggesting nonlinear changes in emotion regulation across adolescence, with decreases in emotion regulation strategies until age 15, followed by increases (e.g., Zimmermann & Iwanski, 2018). Prior studies have identified differing orthogonality of affect at the between- and within-person levels (e.g., Bleidorn & Peters, 2011; Brose, Voelkle, et al., 2015). Our work suggests there may be developmental shifts in how changes in emotions relate to one another. Additionally, these results may provide preliminary evidence suggesting distinct developmental periods during which certain interventions may be most effective. Therapies targeting PA dimensions, for instance, may be more effective when negative

Table 4
Shifts in Indices of Affective Complexity Across Cohorts

Cohort	Dimension	Edge density	Negative edges between PA/NA dimension	R^2	MAE	Entropy
Late childhood (Grade 3)	3	.328	0	.451	.544	4.277
Early adolescence (Grade 6)	4	.320	9	.502	.487	4.160
Middle adolescence (Grade 9)	3	.213	3	.528	.574	3.894

Note. PA = positive affect; NA = negative affect; MAE = mean absolute error.

edges between PA and NA dimensions are strongest and may be less effective when these dimensions are more independent.

Third, we examined possible shifts in dimensionality across cohorts. Whereas many studies of the between-person structure of affect identify a two-factor structure for the PANAS (e.g., Watson & Tellegen, 1985; Wedderhoff et al., 2021), our results suggest greater dimensionality of change in affect for all cohorts. Several reasons for caution in interpreting these findings should be noted, however. First, for early adolescence, we identified a five-dimension network nearly as often as a four-dimension network. Additionally, four-dimension solutions were identified most often for late childhood and middle adolescence due to minor factors, resulting in three-dimension structures being selected instead. Minor factors are generally thought to belong to the broader dimension, in this case a broader PA dimension. Despite these caveats, our results suggest decreasing dimensionality from early to middle adolescence and align more with findings suggesting decreasing emotion differentiation of negative emotions across adolescence (Nook, 2021; Starr et al., 2020). Our results also suggest possible inconsistencies in affect across cohorts, which may indicate that comparisons of PANAS scores between age groups are inappropriate.

Together, these results suggest that different indices of emotional complexity may yield disparate conclusions even within a single study, indicating the importance of additional studies using similar methodologies to clarify normative changes in complexity. At present, there is no single best way to quantify emotional complexity. Our results suggest that these indices provide different perspectives on emotional complexity. In general, we expect global indices of emotional complexity, such as edge density and von Neumann entropy, to better capture the intended meaning of emotional complexity because they consider all emotions, their components, and their interactions in a single, overall metric. Local network characteristics, such as the number of negative edges between dimensions, may reflect complexity on a more specific or nuanced level (e.g., specific emotions). However, further research is needed to clarify the relative strengths of these approaches as measures of emotional complexity, particularly with respect to mental health outcomes. These results together suggest the importance of utilizing longitudinal study designs that can capture changes in emotions over time and across important developmental periods. Such methods may identify more nuanced subdimensions of PA and NA and clarify indices of risk for depression in youth.

Constraints on Generality

Strengths of this study include its accelerated cohort design encompassing childhood and adolescence and its use of rigorous, repeated-measures assessment of affective change across the adolescent transition. This study also used advanced statistical analyses to evaluate youth emotional development in a sophisticated and nuanced fashion.

Several limitations should be noted. The present study measured affect using the PANAS-C, which includes relatively few low-arousal positive emotion states. Future work should evaluate the role of other discrete emotions, such as "gratitude" and "love," within youths' broader emotion networks across development. Additionally, our removal of redundant items from networks may limit the generalizability of our findings, as other studies may identify distinct redundancies. Further, analyses generated undirected edges for each network, and we cannot draw conclusions about the directionality of changes in affect dimensions. Understanding whether increases in dimensions representing fearful emotions precede increases in dimensions representing distressed emotions, for example, is an important question for future research. Additionally, our findings may vary across important populations (e.g., cultural groups, see Raval & Walker, 2019; sex assigned at birth, see Hankin et al., 1998), but we were concerned about the power of such post hoc analyses, so we did not investigate possible differences in this study. Future studies with larger samples that can accommodate analyses using invariance (e.g., Jamison et al., 2024) or trees (e.g., Goretzko & Sterner, 2024) methods may clarify whether these developmental changes differ for distinct populations.

Summary

In this study, we used EGA to examine developmental changes in affect from third through 12th grade. Our findings suggest that different EGA network indices of emotional complexity yield disparate results regarding developmental shifts in complexity. Three indices, edge density, R^2, and von Neumann entropy, suggested decreases in emotional complexity from late childhood through middle adolescence. Negative edges between PA and NA dimensions and MAE, on the other hand, suggested nonlinear shifts in emotional complexity. Last, we observed possible shifts in the dimensionality of affect across cohorts. Our results are exploratory and use novel techniques for estimating longitudinal networks. Further research will be needed to clarify the generalizability of our findings and the clinical utility of these indices. Identifying clinically meaningful differences in indices like entropy, for instance, will be important for comparing findings across groups.

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Received July 17, 2024
Revision received July 25, 2025
Accepted August 19, 2025 ■